Class 23 Global Bifurcations

- Problem set 08 is due Friday.
- There is a project assignment as part of the problem set.

Big picture

Most of the bifurcations we have encountered involve the change in stability of a fixed point. That means there is a straightforward way to identify the bifurcation point (by looking for the parameter value associated with a change of stability).

There are other bifurcations, though, that are harder to detect, because they involve the birth or death of a limit cycle, which is not as easy to notice. Those are the types of bifurcations we are learning about today.

Extra vocabulary / extra facts:

A **bifurcation** is a qualitative change in dynamics that occurs with a small change in a parameter.

Presented last time: remind but don't rewrite

A bifurcation is called **local** if it can be studied via a Taylor series expansion in a small neighborhood of a location in phase space and a point in parameter space. Examples: saddle-node bifurcation, transcritical bifurcation, pitchfork bifurcation, Hopf bifurcation

A **global bifurcation** is intrinsically nonlocal. The **homoclinic bifurcation** is one example, as is a **saddle-node bifurcation of limit cycles**. In this class, we are learning about just a few of these global bifurcations.

	amplitude	period
supercritical Hopf	$\mathcal{O}(\sqrt{\mu})$	$\mathcal{O}(1)$
saddle node of cycles	$\mathcal{O}(1)$	$\mathcal{O}(1)$
infinite period	$\mathcal{O}(1)$	$\mathcal{O}\left(\frac{1}{\sqrt{\mu}}\right)$
homoclinic	$\mathcal{O}(1)$	$\mathcal{O}(\ln \mu)$

Teams 3 and 4: Post screenshots of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

Questions

1. For each of the following bifurcations, answer the following:
   - What is the minimum number of fixed points you expect in the phase portrait?
   - What kind of fixed point(s) could there be?
   - Does the number or type of fixed points change during the bifurcation?
   - How will the number of stable, and of unstable, limit cycles change during the bifurcation?

   (a) saddle-node of limit cycles

   (b) infinite-period

   (c) homoclinic

Answers:

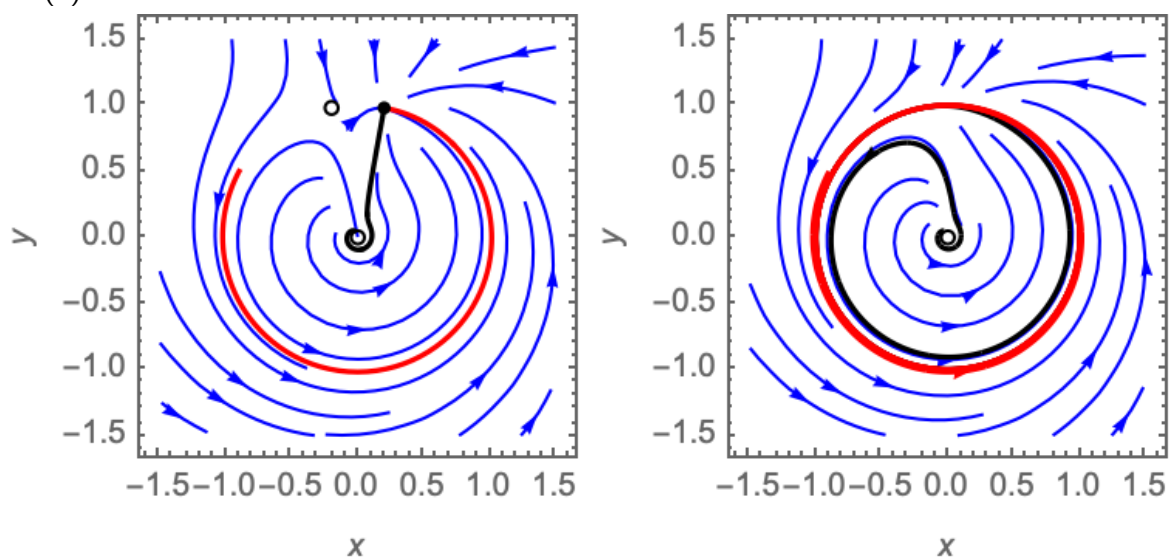
a. 1 fixed point (the limit cycles surround a set of fixed points with index $+1$). It would be a repeller or attractor. Nothing happens to the fixed point during the bifurcation. One stable and one unstable limit cycle is lost (each goes down by 1).

b: 3 fixed points on one side and one on the other. It would be a saddle and a node (attractor or repeller) along with a fixed point at the center of the limit cycle (an attractor or repeller). A fixed point is born at the moment of bifurcation (a saddle-node bifurcation coincides with a limit cycle). One stable limit cycle is lost in the bifurcation.

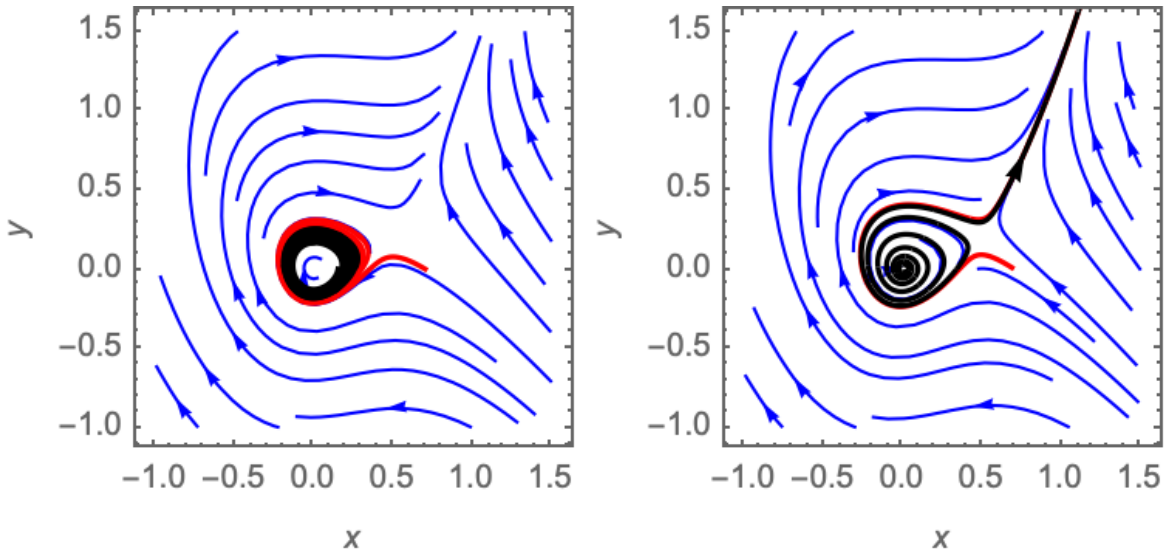
c: 2 fixed points (the limit cycle surrounds a set of fixed points with index $+1$, and there is a saddle point). The fixed point inside the limit cycle would be a repeller or attractor. The other fixed point is a saddle point. Nothing happens to the fixed points during the bifurcation. One stable limit cycle is lost.

2. Each pair of phase portraits below are for a system on either side of a bifurcation. For each pair of phase portraits, fill in the chart and identify the bifurcation.

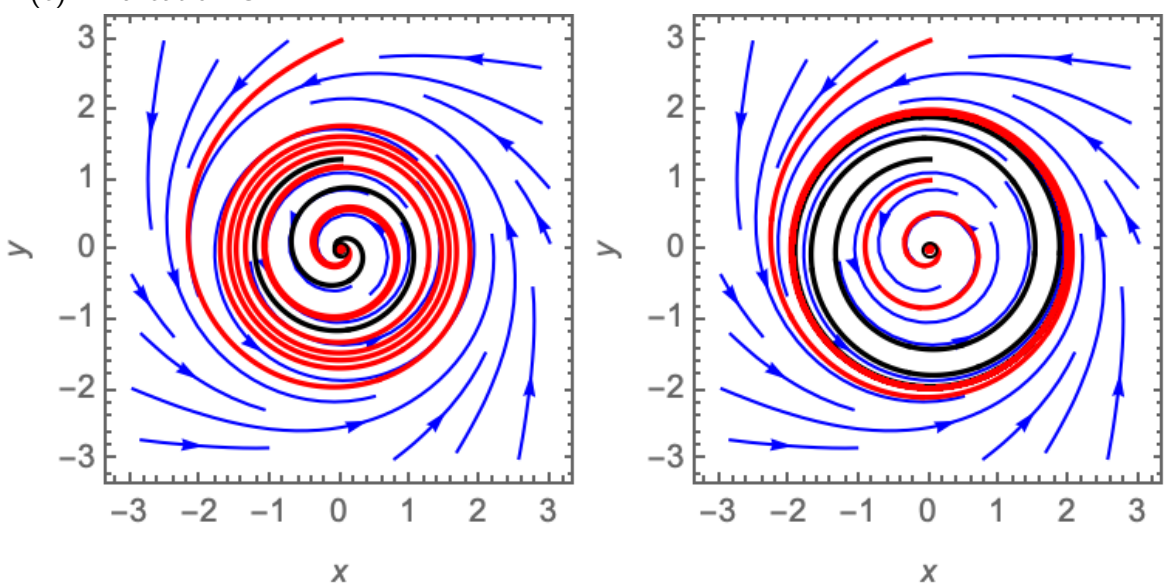
   (a) Bifurcation A



   (b) Bifurcation B



(c) Bifurcation C



	left	right
# of fixed points		
# of stable limit cycles		
# of unstable limit cycles		

Bifurcation:

◯ saddle node of cycles ◯ saddle node infinite period ◯ homoclinic

Answers

(a) This is a saddle node infinite period bifurcation. On the left there are three fixed points: one repeller, one attractor, and one saddle point. THere are no limit cycles. On the right there is an attracting limit cycle and a repelling fixed point.

(b) This is a homoclinic bifurcation. On the left there are two fixed points, a repeller and a saddle point. There is also an attracting limit cycle. On the right there are the same two fixed points and no limit cycles.

(c) This is a saddle node of limit cycles bifurcation. On the left there is an attracting fixed point and no limit cycles (flows from the outside edge spiral all the way in). On the right there is a pair of limit cycles, one attracting and one repelling, along with the attracting fixed point.

3. (Timing of bifurcations)

(a) We have encountered the infinite-period bifurcation before, when we studied flows on the circle. In Class 10 we computed how long it takes to flow around the circle as we approached the bifurcation. We found that the time was proportional to $\frac{1}{\sqrt{r}}$, where r was the distance from the bifurcation in parameter space.

How is the bifurcation we studied in Class 10 related to the infinite-period bifurcation we are seeing in 2D?

(b) In a homoclinic bifurcation, a limit cycle approaches a saddle point. As the limit cycle becomes close to the saddle point, the time it takes to traverse the cycle is dominated by the time spent near the saddle point. Why is that?

(c) We can calculate the time spent near the saddle point using linearization. Let μ be the distance of the limit cycle from the stable manifold of the saddle point (measured along the unstable direction), at a distance k up the stable manifold from the saddle point.

Assume that the limit cycle is close enough to the saddle point that a linear approximation is appropriate to describe the dynamics around the saddle point.

In that case, solutions are $c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2$ with $\lambda_1 < 0$ and $\lambda_2 > 0$. Let $\mathbf{v}_1$ and $\mathbf{v}_2$ each be length 1.

Choose c_1 and c_2 to match the scenario described. *One will be set to μ and the other to k.*

(d) A particle starting at the point above will flow past the saddle point to k distance along the unstable manifold when $c_i e^{\lambda_i T} = k$ (where you have determined whether $i = 1$ or $i = 2$ in your work above). Find an expression for T, the time when this occurs. How does this time depend on μ?

Answers

(a) The flow on the circle is like the flow around the limit cycle. In both cases the saddle-node bifurcation happens to disrupt the cycling.

(b) The vector field is slow near the saddle point and the closer you get to the saddle point the slower it is.

(c) The stable direction is $\mathbf{v}_1$ and the unstable direction is $\mathbf{v}_2$. So $c_2 = \mu$ and $c_1 = k$.

(d) We have $\mu e^{\lambda_2 T} = k$ so $e^{\lambda_2 T} = \frac{k}{\mu}$ and $T = \frac{1}{\lambda_2}\ln\left(\frac{k}{\mu}\right) = \frac{1}{\lambda_2}\ln k - \frac{1}{\lambda_2}\ln\mu$. The dependence on μ is that the time is some constant plus a quantity proportional to $\ln\mu$

4. **1D map cobweb example**. $x_{n+1} = f(x_n)$ with $f(x) = 2x(1-x)$.

From left to right:

1. Start the orbit (and cobweb) at x_0 on the horizontal axis.

2. Draw a vertical line connecting x_0 to $x_1 = f(x_0)$. Find x_1 along the horizontal axis by drawing a horizontal line (x_0, x_1) to (x_1, x_1).

   x_0 and x_1 are marked along the horizontal axis.

   The orbit lives in 1d. The pair (x_0, x_1) represents two consecutive points in the orbit.

3. Draw a vertical line from x_1 up to $x_2 = f(x_1)$. Find x_2 along the horizontal axis by drawing a horizontal line (x_1, x_2) to (x_2, x_2).

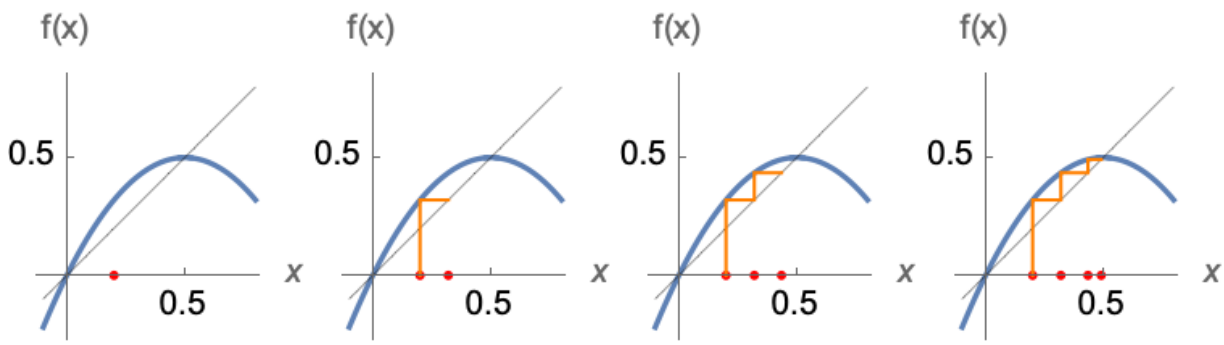
   x_0, x_1, x_2 are marked along the horizontal axis.

   The orbit lives in 1d. The pair (x_1, x_2) represents two consecutive points in the orbit.

4. Draw a vertical line from x_2 up to $x_3 = f(x_2)$. Find x_3 along the horizontal axis by drawing a horizontal line (x_2, x_3) to (x_3, x_3).

   x_0, x_1, x_2, x_3 are marked along the horizontal axis.

   The orbit lives in 1d. The pair (x_2, x_3) represents two consecutive points in the orbit.



Practice.

Let $x_{n+1} = f(x_n)$ with $f(x) = \cos x$.

(a) Plot $\cos x$ and x on the same plot.

(b) Use your plot to identify any fixed points of the map.

(c) Make cobwebs for at least two qualitatively different cases.